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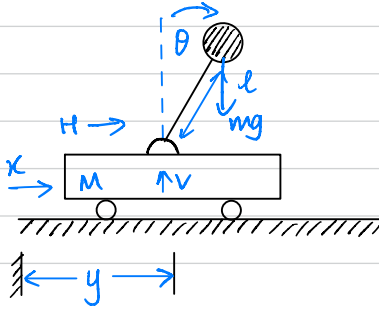
RMC Final Exam

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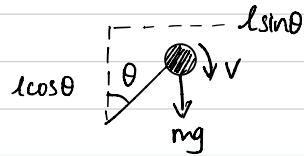
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1)



a) Free Body Diagram:



Applying Lagranges Approach; $T = \frac{1}{2} m(\dot{y})^2 + \frac{1}{2} m(R\dot{\theta} + \dot{y})^2$

$$\Rightarrow T = \frac{1}{2} m \dot{y}^2 + \frac{1}{2} m (L^2 \dot{\theta}^2 + \dot{y}^2 + 2L\dot{\theta}\dot{y}\cos\theta)$$

$$V = mgl\cos\theta$$

$$W = U + V$$

$$L = T - V$$

$$L = \frac{1}{2} m \dot{y}^2 + \frac{1}{2} m [L^2 \dot{\theta}^2 + \dot{y}^2 + 2L\dot{\theta}\dot{y}\cos\theta] - mgl\cos\theta$$

$$\left\{ \text{as } \frac{d}{dt} \left[\frac{\partial L}{\partial \dot{\theta}} \right] - \frac{\partial L}{\partial \theta} = 0 \right\}$$

$$mL^2\ddot{\theta} + m\dot{y}L\cos\theta - m\dot{y}L\sin\theta + m\dot{y}L\dot{\theta}\sin\theta - mgl\sin\theta = 0.$$

$$\Rightarrow mL^2\ddot{\theta} + m\dot{y}L\cos\theta - mgl\sin\theta = 0.$$

$$\Rightarrow L\ddot{\theta} + \dot{y}\cos\theta - g\sin\theta = 0.$$

Hence the Dynamic equations are,

$$(M+m)\ddot{y} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = u$$

$$ml^2\ddot{\theta} + m\dot{y}l\cos\theta = mgl\sin\theta$$

It is clear that the Dynamic equations are non-linear because they consist of Trigonometric quantities.

b) By assuming θ is very small,

$$\text{i.e. } \cos\theta \approx 1$$

$$\sin\theta \approx \theta$$

$$\theta^2 \approx 0.$$

The Dynamic equation can be rewritten as:

$$(M+m)\ddot{y} + ml\ddot{\theta}(1) - ml(0)\theta = u.$$

$$\therefore (M+m)\ddot{y} + ml\ddot{\theta} = u$$

$$ml^2\ddot{\theta} + ml\dot{y}(1) = mgl\theta$$

$$\therefore ml^2\ddot{\theta} + ml\dot{y} = mgl\theta$$

Thus, if we assume θ is very small, the Dynamic equations can be represented by two coupled linear Differential equations.

c) To find $\frac{\theta(s)}{u(s)}$:

$$(M+m)\ddot{y} + ml\ddot{\theta} = u$$

$$s^2(M+m)y(s) + s^2ml\theta(s) = u(s) \quad \text{--- (i)}$$

$$l\ddot{\theta} + \dot{y} = g\theta$$

page -3

$$s^2 l \theta(s) + s^2 y(s) = g \theta(s)$$
$$(s^2 l - g) \theta(s) = -s^2 y(s)$$

$$\theta(s) = \frac{-s^2 y(s)}{s^2 l - g} \quad \text{--- (ii')}$$

$$y(s) = \frac{g - s^2 l \theta(s)}{s^2} \quad \text{--- (iii')}$$

By putting (iii') in (i) we obtain,

$$\frac{s^2 (M+m) (g - s^2 l) \theta(s)}{s^2} + s^2 m l \theta(s) = u(s)$$

$$\theta(s) [g(M+m) - s^2 l(M+m) + s^2 m l] = u(s)$$

$$\frac{\theta(s)}{u(s)} = \frac{1}{g(M+m) - s^2 l(M+m - m)}$$

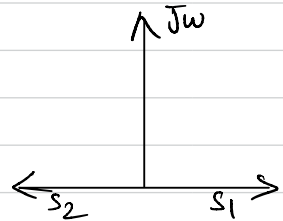
$$\frac{\theta(s)}{u(s)} = \frac{1}{g(M+m) - s^2 L M}$$

To find poles,

$$g(M+m) - s^2 m l = 0$$

$$s^2 = \frac{g}{m l} (M+m)$$

$$s_{1,2} = \pm \sqrt{\frac{(M+m)g}{m l}}$$



we can see the one pole is on the Right side of the plane, so the system is unstable.

Page-4

To find $y(s)/u(s)$:
put ② in ①,

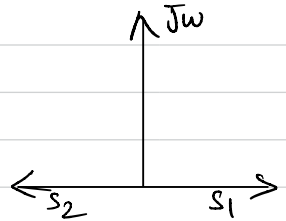
$$s^2(M+m)y(s) + \frac{s^2 ml(-s^2 y(s))}{s^2 l - g} = u(s)$$

$$s^2(s^2 l - g)(M+m)y(s) + s^2 ml(-s^2 y(s)) = (s^2 l - g)u(s)$$
$$y(s) [s^4 l(M+m) - s^4 ml - s^2 g(M+m)] = (s^2 l - g)u(s)$$

$$\therefore \frac{y(s)}{u(s)} = \frac{s^2 l - g}{s^4 l(M+m) - s^4 ml - s^2 g(M+m)}$$

To find poles,

$$s^4 ml - s^2 g(M+m) = 0$$
$$s^4 ml = s^2 g(M+m)$$
$$s^2 = (g/ml)(M+m)$$
$$s_{1,2} = \pm \sqrt{(g/ml)(M+m)}$$



As one pole is on the right side of plane, system is unstable.

d)

$$\theta = x_1$$
$$\dot{\theta} = \dot{x}_1 = x_2$$
$$y = x_3$$
$$\dot{y} = \dot{x}_3 = x_4$$

page-5

$$(M+m)\ddot{y} + ml\ddot{\theta} = f$$

$$ml\ddot{\theta} = mg\theta - m\ddot{y}$$

$$\ddot{\theta} = \frac{mg\theta - m\ddot{y}}{ml} \quad \text{--- (1)}$$

$$(M+m)\ddot{y} + mg\theta - m\ddot{y} = u$$

$$M\ddot{y} + mg\theta = u$$

$$\ddot{y} = \frac{u}{M} - \frac{mg\theta}{m} \quad \text{--- (2)}$$

putting (2) in (1),

$$\ddot{\theta} = \frac{mg\theta}{ml} - \frac{1}{l} \left(\frac{u}{M} - \frac{mg\theta}{m} \right)$$

$$= \frac{mg\theta}{ml} - \frac{u}{Ml} + \frac{mg\theta}{Ml}$$

$$= \frac{-u}{Ml} + \frac{(M+m)g\theta}{Ml} \quad \text{--- (3)}$$

$$\dot{x}_4 = \ddot{y} = \frac{u}{M} - \frac{mg}{M} x_1$$

$$\dot{x}_2 = \ddot{\theta} = \frac{-u}{Ml} + \frac{(M+m)g}{Ml} x_1$$

page-6

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{(M+m)g}{ML} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{mg}{M} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{1}{ML} \\ 0 \\ \frac{1}{M} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

e) For state Feedback control,

$$|\lambda I - (A+BK)| = \Delta d$$

comparing with state control law using MATLAB,

$$K = [-56.8242, -12.5484, -2.4465, -5.0968]$$

MATLAB code is attached in the next page:

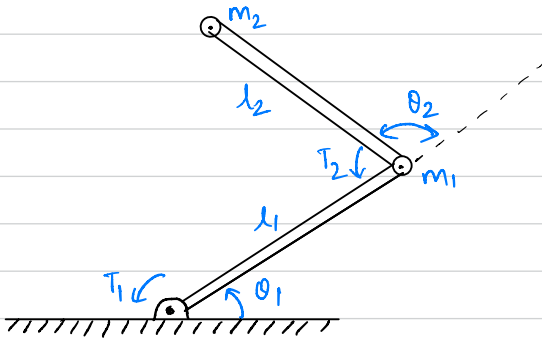
page-7

```
1 M = 2;  
2 m = 0.1;  
3 l = 0.5;  
4 g = 9.81;  
5  
6 A = [0 1 0 0; (M+m)*g/(M*l) 0 0 0; 0 0 0 1; -(m*g)/M 0 0 0];  
7 B = [0; -1/(M*l); 0; 1/M];  
8 C = [1 0 0 0; 0 0 1 0];  
9 D = 0;  
10  
11 p1 = -1;  
12 p2 = -2;  
13 p3 = -3;  
14 p4 = -4;  
15 |  
16 K = place(A, B, [p1 p2 p3 p4])
```

The output received is as follows:

$$K = [-56.8242, -12.5484, -2.4465, -5.0968]$$

2)



a) Jacobian of two link manipulator,

$$J = \begin{bmatrix} -L_1 S_1 - L_2 S_{12} & -L_2 S_{12} \\ L_1 C_1 + L_2 C_{12} & L_2 C_{12} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

Case-I:

$$\dot{\theta}_1 = 1 \text{ rad/sec}^{-1}, \dot{\theta}_2 = 3 \text{ rad/sec}^{-1}$$

$$\theta_1 = 60^\circ, l_1 = 2 \text{ m}, \theta_2 = 30^\circ, l_2 = 3 \text{ m}.$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -2 \sin 60^\circ - 3 \sin 90^\circ & -3 \sin 90^\circ \\ 2 \cos 60^\circ + 3 \cos 90^\circ & 3 \cos 90^\circ \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -\sqrt{3} - \sqrt{3} & -3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -\sqrt{3} - 3 - 9 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -13.732 \\ 1 \end{bmatrix}$$

Page -9

case - II :

$$\theta_1 = 167^\circ, \quad l_1 = 2\text{m}$$

$$\theta_2 = -156^\circ, \quad l_2 = 3\text{m}$$

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} -2\sin(167^\circ) - 3\sin(-156^\circ) & -3\sin(-156^\circ) \\ 2\cos(167^\circ) + 3\cos(-156^\circ) & 3\cos(156^\circ) \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} -1.0223 & -0.5724 \\ 0.9961 & 2.9449 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} -2.7396 \\ 9.8308 \end{bmatrix} \text{ m/s}^2$$

b) For $F = \begin{bmatrix} 30 \\ 20 \end{bmatrix}$ $Z = J^T F$

case - I :

$$Z = \begin{bmatrix} -4.732 & 1 \\ -3 & 0 \end{bmatrix} \begin{bmatrix} 30 \\ 20 \end{bmatrix} \quad \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} = \begin{bmatrix} -121.96 \\ -90 \end{bmatrix} \text{ Nm}$$

case - II :

$$Z = \begin{bmatrix} -1.022 & 0.9961 \\ -0.5724 & 2.9449 \end{bmatrix} \begin{bmatrix} 30 \\ 20 \end{bmatrix} \quad \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} = \begin{bmatrix} -10.738 \\ 41.7248 \end{bmatrix} \text{ Nm}$$

$$\text{For } F = \begin{bmatrix} 30 \\ 20 \end{bmatrix}$$

case I:

$$Z = \begin{bmatrix} -3-\sqrt{3} & 1 \\ -3 & 0 \end{bmatrix} \begin{bmatrix} 30 \\ -20 \end{bmatrix} \quad \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} -161.9615 \\ -90 \end{bmatrix} \text{ Nm}$$

case-II:

$$Z = \begin{bmatrix} -1.0223 & 0.9961 \\ -0.5724 & 2.9449 \end{bmatrix} \begin{bmatrix} 30 \\ -20 \end{bmatrix} \quad \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} -50.59 \\ -76.07 \end{bmatrix} \text{ Nm}$$

If we compare the results, we can say that the intended force at the tip will depend on the arm design and will change accordingly for its configuration. The magnitude and direction torques will follow the parameters & configurations as per mentioned.

c) Dynamic equation of two-link manipulator can be given by,

$$Z = M(\theta)\ddot{\theta} + \gamma(\theta, \dot{\theta}) + G(\theta)$$

$$M(\theta) = \begin{bmatrix} l_2^2 m_2 + 2l_1 l_2 m_2 c_2 + l_1^2 (m_1 + m_2) & l_2^2 m_2 + l_1 l_2 m_2 c_2 \\ l_2^2 m_2 + l_1 l_2 m_2 c_2 & l_2^2 m_2 \end{bmatrix}$$

$$\gamma(\theta, \dot{\theta}) = \begin{bmatrix} -m_2 l_1 l_2 s_2 \dot{\theta}_2^2 - 2m_2 l_1 l_2 s_2 \dot{\theta}_1 \dot{\theta}_2 \\ m_2 l_1 l_2 s_2 \dot{\theta}_1^2 \end{bmatrix}$$

page -11

$$G(\theta) = \begin{bmatrix} m_2 l_2 g c_{12} + (m_1 + m_2) l_1 g c_1 \\ m_2 l_2 g c_{12} \end{bmatrix}$$

According to control law partitioning law,

$$z = m\ddot{\theta} + v\dot{\theta} + G\theta$$

$$z = \alpha z' + \beta$$

Let $\alpha = M$, $\beta = v\dot{\theta} + G\theta$; $\ddot{\theta} = z'$

For stabilizing the system,

$$z' = -K_v \dot{\theta} - K_p \theta$$

$$\ddot{\theta} + K_v \dot{\theta} + K_p \theta = 0 \quad \text{--- (1)}$$

$$K_v = 2\sqrt{K_p}$$

According to (1),

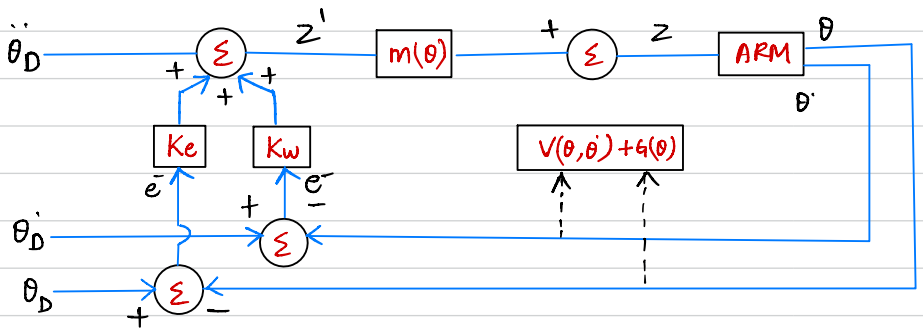
$$s^2 + K_v s + K_p = s^2 + 2\omega_n s + \omega_n^2$$

thus,

$$K_v = K_w = \begin{bmatrix} K_{w1} & 0 \\ 0 & K_{w2} \end{bmatrix}$$

$$K_p = K_\theta = \begin{bmatrix} K_{\theta 1} & 0 \\ 0 & K_{\theta 2} \end{bmatrix}$$

for critical damping, $K_{wi} = 2\sqrt{K_i}$



3)

Given,

$$\tau = ml^2 \ddot{\theta} + f \dot{\theta} + mgl \cos \theta$$

a) $w_n = 10, m = 1, l = 1, f = 7, g = 10$

$$\Rightarrow \tau = \ddot{\theta} + 7\dot{\theta} + 10 \cos \theta$$

Method of control law partitioning,

(i) Model based portion:

$$\ddot{\theta} + 7\dot{\theta} + 10 \cos \theta = \tau$$

$$\tau = \alpha \tau' + \beta$$

$$\Rightarrow \ddot{\theta} + 7\dot{\theta} + 10 \cos \theta = \alpha \tau' + \beta$$

(ii) Servo portion:

$$\alpha = m = 1, \quad \beta = f \dot{x} + kx = 7\dot{\theta} + 10 \cos \theta$$

$$\Rightarrow \ddot{\theta} + 7\dot{\theta} + 10 \cos \theta = \tau' + 7\dot{\theta} + 10 \cos \theta$$

$$\Rightarrow \ddot{\theta} = \tau'$$

$$\tau^1 = -K_v \dot{\theta} - K_p \theta + r \quad (r=0, \text{ is the reference})$$

$$\tau^1 + K_v \dot{\theta} + K_p \theta = 0 \longrightarrow \ddot{\theta} + K_v \dot{\theta} + K_p \theta = 0$$

$$s^2 + K_v s + K_p = 0$$

$$s^2 + K_v s + K_p \equiv s^2 + 2\zeta\omega_n s + \omega_n^2$$

$\zeta = 1 \rightarrow$ for critically damped

$\omega_n = 10 \rightarrow$ given

$$\Rightarrow s^2 + K_v s + K_p \equiv s^2 + 2(1)(10)s + 10^2$$

$$\{ K_v = 20, K_p = 100 \}$$

b) To Track the desired Trajectory,

$$\theta_d = A \sin(2\pi t / T)$$

$$A = 0.1 \text{ rad (6 degrees)}$$

$$T = 2 \text{ sec}$$

$$\tau^1 = \ddot{\theta}_d + K_v \dot{E} + K_p E$$

Error equations:

$$E = \theta_d - \theta$$

$$\dot{E} = \dot{\theta}_d - \dot{\theta}$$

$$\ddot{E} = \ddot{\theta}_d - \ddot{\theta}$$

$$\theta_d = 0.1 \sin\left(\frac{2\pi t}{2}\right)$$

$$\theta_d = 0.1 \sin(\pi t)$$

$$\dot{\theta}_d = 0.1 \pi \cos(\pi t) = 0.314 \cos(\pi t)$$

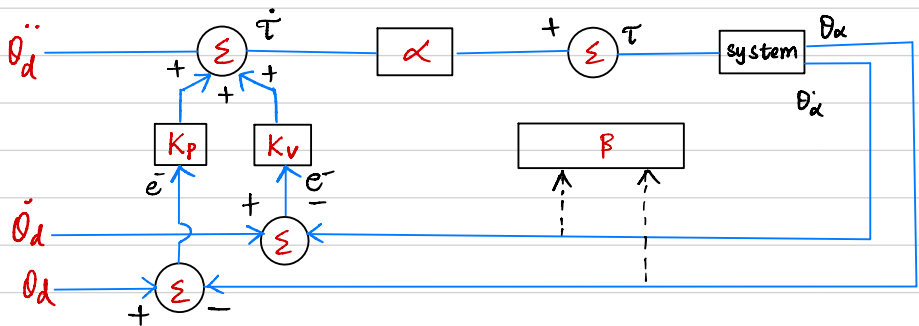
$$\ddot{\theta}_d = -0.1 \pi^2 \sin(\pi t) = -0.986 \sin(\pi t)$$

$$K_p = 100 \quad K_v = 20$$

$$T' = -0.986 \sin(\pi t) + 20\dot{E} + 100E$$

(or)

$$= -0.1 \pi^2 \sin(\pi t) + 20\dot{E} + 100E$$



c)

$$T = ml^2 \ddot{\theta} + f \dot{\theta} + mgl \cos \theta$$

put values $m=1, l=1, f=7, g=10$

$$T = \ddot{\theta} + 7\dot{\theta} + 10 \cos \theta$$

using standard rotations $T = u, \theta \rightarrow y$

$$u = \ddot{y} + 7\dot{y} + 10 \cos y$$

for state space eqn,

$$x_1 = y$$

$$\dot{x}_2 = \dot{y} - x_1$$

$$\ddot{x}_2 = \ddot{y}$$

$$\text{then, } \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ 4 - 7x_2 - 10x_1 \end{bmatrix} = f(x, u) \quad \text{--- (1)}$$

$$y = x_1 - g(x_1, u) \quad \text{--- ②}$$

Finding equilibrium point (x_0, u_0)
 let's take

$$x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

to find $f(x_0, u_0) = 0$

$$f(x_0, u_0) = \begin{bmatrix} 0 \\ u_0 - 10 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow u_0 = 10.$$

To find state space matrices,

$$A = \frac{\partial f}{\partial x}(x_0, u_0) = \begin{bmatrix} 0 & 1 \\ 10 \sin 0 & -7 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -7 \end{bmatrix}$$

$$B = \frac{\partial f}{\partial u}(x_0, u_0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \frac{\partial g}{\partial x}(x_0, u_0) = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \frac{\partial g}{\partial u}(x_0, u_0) = \begin{bmatrix} 0 \end{bmatrix}$$

$$\left. \begin{aligned} \Delta x &= \begin{bmatrix} 0 & 1 \\ 0 & -7 \end{bmatrix} \Delta x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Delta u \\ \Delta y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \Delta x \end{aligned} \right\} \quad \text{--- ①}$$

Page - 16

$$x = x_0 + \Delta x = [0 \ 0]^{-1} + \Delta x$$

$$u = u_0 + \Delta u = 10 + \Delta u$$

$$y = y_0 + \Delta y = 0 + \Delta y = \Delta y$$

From ①, finding Eigen Values for stability

$$(A - \lambda I) = 0$$

$$\Rightarrow \begin{bmatrix} \lambda & 1 \\ 0 & -7-\lambda \end{bmatrix} = 0$$

$$\rightarrow -\lambda \cdot (7+\lambda) = 0$$

$$\rightarrow \lambda = 0 \text{ \& } \lambda = -7$$

We can see that one pole is at 0.

So, the system should be unstable.